TEXORA · LECTURE SHEET

Basic Concept of Dyes and Pigments

Solubility and affinity define a dye; insolubility and a binder define a pigment — and the seven dye–fibre forces explain every recipe and fastness grade that follows.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 2 · LEVEL 3 TERM 1

1. Dye or pigment? The defining test

A **dye** is applied from a medium in which it is soluble (or is made temporarily soluble), has substantivity for the fibre, and is fixed at molecular level inside the fibre. A **pigment** is insoluble in the application medium and in the fibre; it is held on the surface by a polymer **binder** film. The test is solubility plus affinity: disperse dyes are borderline — sparingly soluble, applied as fine dispersions, dissolving molecularly into polyester. Pigments are covered fully in Topics 4 and 10; this sheet builds the dye side and the forces that hold any colorant to a fibre.

2. What a commercial dye must do

- Exhaust from the bath onto the fibre at a controllable rate; build up to the target depth reproducibly.
- Fix or be retained with adequate fastness for the end use; unfixed dye must wash out cleanly.
- Behave levelly: migrate enough to cover fibre irregularities, yet not so mobile that fastness suffers.
- Survive the process window (temperature, pH, electrolytes, oxidants/reductants) without shade change.
- Meet cost, effluent and restricted-substance (e.g. banned azo amines) requirements.

3. The forces between dye and fibre

FORCE	NATURE	WHERE IT DOMINATES
Covalent bond	Shared-electron chemical bond, strongest (~200–400 kJ/mol)	Reactive dyes on cellulose, wool, silk, nylon
Ionic (salt) linkage	Attraction of dye anions to protonated fibre sites, or dye cations to anionic sites	Acid dyes on wool/nylon ($–NH_3^+$); basic dyes on acrylic ($–SO_3^-$/$–COO^-$)

Hydrogen bonds	Dye –OH/–NH donors to fibre O/N acceptors (~10–30 kJ/mol each, numerous)	Direct dyes on cellulose; disperse dyes on acetate
van der Waals / hydrophobic	Short-range attraction maximised by long, planar, coplanar molecules; water exclusion	Direct dyes (long planar molecules lying along cellulose chains); disperse dyes in polyester
Coordination (chelation)	Metal ion bridging dye and fibre, or pre-metallised dye complex	Mordant dyes; 1:1 and 1:2 metal-complex acid dyes
Mechanical entrapment	Insoluble pigment formed inside the fibre pores	Vat and sulphur dyes (oxidised back inside), azoic (naphthol) combinations
Solid solution	Dye dissolves in the fibre polymer itself	Disperse dyes in polyester and acetate

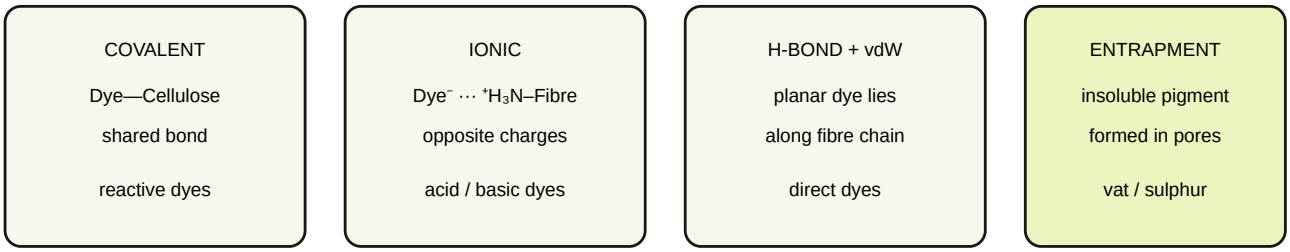


Fig. 1 — Four principal ways a dye is held by a fibre, from strongest (covalent) to purely physical (entrapment).

4. The dye classes at a glance

CLASS	FIBRE	APPLICATION CHARACTER	FASTNESS PROFILE
Direct	Cotton, viscose	Anionic, from neutral/alkaline bath with salt	Light moderate; wash poor–moderate (improved by after-treatment)
Reactive	Cotton, viscose, wool, silk	Anionic; alkali fixes covalently	Wash excellent; light good
Vat	Cotton	Insoluble; reduced to soluble leuco form, oxidised back in fibre	Excellent all round

Sulphur	Cotton	Reduced with sulphide; cheap dark shades	Good wash; poor chlorine/rub; tendering risk on storage
Azoic	Cotton	Naphthol + diazonium coupled inside fibre	Brilliant reds; rub fastness limited
Acid	Wool, silk, nylon	Anionic, acid bath; levelling → milling → metal-complex grades	Rises with molecular size/complexity; wash good in milling grades
Basic	Acrylic, modacrylic	Cationic; brilliant shades	Light poor on natural fibres; good on acrylic
Disperse	Polyester, acetate, nylon	Non-ionic dispersion; high temperature or carrier	Wash excellent; sublimation varies by energy class
Mordant	Wool, cotton	Applied with metal mordant (Cr, Al, Fe)	Excellent; chrome effluent restricts use

5. Pigments — the contrasting system

Pigments (organic: azo, phthalocyanine; inorganic: titanium dioxide white, carbon black, iron oxides) have no fibre affinity at all. In pigment dyeing and printing the pigment, binder, thickener and auxiliaries are padded or printed, dried and cured; the crosslinked binder film is the entire fastness story — wash fastness is good while the film holds, rub fastness and handle are the weak points because colour sits on the surface. Pigments work on *any* fibre and blend uniformly, which is why they dominate printing (Topic 10) and garment dyeing effects.

6. Choosing a class — the decision logic

Start from the fibre (it dictates which forces are available), then impose the fastness specification, shade brilliance and cost. Cotton for apparel with high wash fastness → reactive or vat; economy dark workwear → sulphur; polyester sportswear → disperse HTHP; wool suiting → metal-complex acid or reactive-for-wool; acrylic knitwear in brilliant shades → basic. TE-3106 Experiment 2 applies four of these classes side by side so the logic is seen on real fabric.

7. Glossary — key terms

- **Substantivity:** the inherent attraction of a dissolved dye for the fibre from a bath.
- **Exhaustion:** the percentage of bath dye that transfers to the fibre.
- **Binder:** film-forming polymer that fixes pigments mechanically to the surface.

- **Leuco form:** the reduced, soluble form of a vat or sulphur dye.
- **Mordant:** a metal salt that bridges dye and fibre by coordination.

8. Key points

- Dye = soluble + substantive + molecularly fixed; pigment = insoluble + binder-held.
- Seven retention mechanisms exist; covalent (reactive) and entrapment (vat) give the highest wash fastness.
- Class selection is fibre-first, then fastness, brilliance and cost.

9. Summary

Dyes and pigments solve the same problem — putting stable colour into a textile — by opposite strategies: molecular dissolution and affinity versus surface filming. The seven dye–fibre forces explain both application recipes and fastness grades, and the class table built on them is the selection chart used for every recipe in this course.

10. Practice

1. A buyer demands wash fastness grade 4–5 on cotton shirting in a bright turquoise. Choose between direct, reactive and sulphur dyes and justify by binding force.
2. Why can one pigment recipe colour a polyester/cotton blend levelly when no single dye can?
3. Disperse dye in polyester is often called a solid solution. What evidence from its behaviour supports that description?
4. Explain why increasing the planarity and length of a direct dye molecule raises substantivity for cellulose.

Answer key

1. Reactive: only it bonds covalently to cellulose, so desorption in washing is minimal; phthalocyanine reactive turquoise gives the brilliance. Direct dyes desorb (H-bonds/vdW only) and sulphur dyes cannot give a bright turquoise.
2. The pigment has no fibre affinity to differ between components; the binder films both fibres identically, so depth is uniform regardless of blend ratio.
3. It dissolves molecularly in the polymer (uptake proportional to solubility in the fibre phase), diffuses as single molecules, and can migrate/sublime out again on heating — solution-like behaviour, not surface adsorption saturation alone.

4. Cellulose chains present extended flat surfaces of –OH groups. A long planar dye maximises contact area for van der Waals forces and aligns multiple –OH/azo groups for simultaneous H-bonds; total attraction is the sum over the whole contact length, so substantivity rises steeply.
